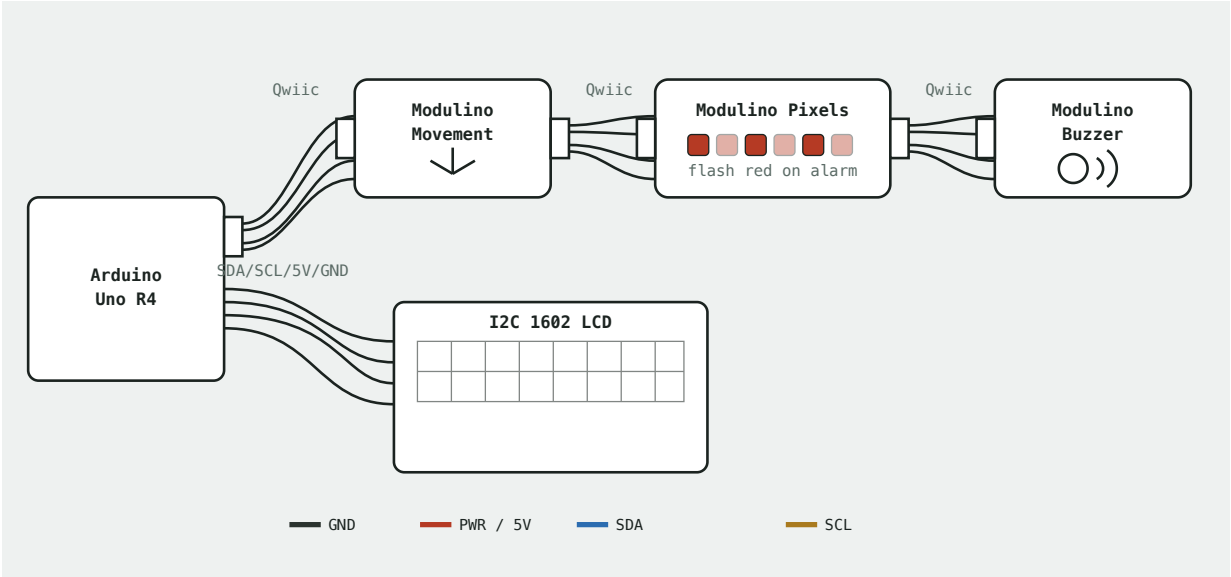
Shake alarm

A motion-sensing alarm. The Modulino Movement module watches for a shake; a shake trips a flashing, beeping alarm for a few seconds, then the alarm re-arms itself. No extra parts beyond the LCD from project 2.

Board · Arduino Uno R4 **Modules** · Movement, Pixels, Buzzer, I2C LCD
New concepts · shake detection, state machines



Three Modulino modules chain on Qwiic as usual; the LCD is wired exactly as in **project 2**. The Pixels glow dim blue while armed and flash red during the alarm.

PIN & CONNECTION REFERENCE

Component	Connects to
Modulino Movement	Qwiic/I2C bus
Modulino Pixels	Qwiic/I2C bus
Modulino Buzzer	Qwiic/I2C bus
I2C 1602 LCD	VCC→5V, GND→GND, SDA→SDA, SCL→SCL (same as project 2)

Gotcha: if the alarm trips too easily (or not at all), change `SHAKE_THRESHOLD` in the sketch — it's currently 80; a bigger number needs a harder shake.

REQUIRED LIBRARIES

- `Arduino_Modulino` — pulls in the sensor-vendor dependencies automatically
- `LiquidCrystal_I2C` — Library Manager

CODE REFERENCE

Detecting a shake

Add up the rotation rate on all three axes; direction doesn't matter.

```
// --- Inside loop() ---
movement.update();
float motion = abs(movement.getRoll())
             + abs(movement.getPitch())
             + abs(movement.getYaw());
bool shaking = motion > SHAKE_THRESHOLD;
```

A small state machine

Naming the modes makes the logic easier to follow than a chain of flags.

```
// --- Variables (top of sketch) ---
enum AlarmState { ARMED, ALARM };
AlarmState state = ARMED;
unsigned long alarmStartTime = 0;

// --- Inside loop() ---
if (state == ALARM) {
  if (millis() - alarmStartTime
      >= ALARM_DURATION_MS)
    state = ARMED;
} else if (motion > SHAKE_THRESHOLD) {
  state = ALARM;
  alarmStartTime = millis();
}
```

TRY IT YOURSELF

- Change `SHAKE_THRESHOLD` to make the alarm more or less sensitive.
- Add a third state, `DISARMED`, and use a Modulino Buttons press to switch the alarm on and off.
- Show how many times the alarm has been triggered on the LCD's second line.